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Machine Learning–Based Food Demand Prediction and Waste Analytics for Buffet Restaurants

Design, Evaluation, and Implementation of a Regression and Cloud-Based Decision Support System

Introduction to Machine Learning

Bachelor of Engineering in Information and Communication Technology (Robotics)

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Akther Rubina, K M Arafat Islam

Md Jahidul Islam, Prova Zarin

HAMK
Häme University of Applied Sciences

Abstract

DP	Bachelor of Engineering in Information and Communication Technology (Robotics)
Members	Akther Rubina, K M Arafat Islam, Md Jahidul Islam, Prova Zarin Year 2026
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Coordinator	Mehnaz Khan

Food waste in buffet restaurants represents a significant operational, economic, and environmental challenge. Inaccurate demand estimation leads to food overproduction and waste, whereas underproduction reduces customer satisfaction and operational efficiency. The objective of this thesis was to design and evaluate a machine learning–based decision support system for predicting next-day food demand and analysing food waste in buffet restaurant environments.

A prototype system, named MIXX, was developed using synthetic restaurant data representing daily dish-level operations. The system predicts next-day demand using historical consumption patterns, temperature data, and holiday indicators as input features. Linear Regression was selected as the primary model due to its interpretability and suitability for structured datasets with limited size. For robustness, a fallback mechanism was implemented to handle new dishes without historical data.

The system architecture integrates Google Colab for model development, BigQuery as a cloud-based data warehouse, and Looker Studio for interactive dashboard visualization. In addition to demand forecasting, the system computes food waste quantity and waste percentage using daily sold and cooked input data.

Model performance was evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). A comparative analysis between Linear Regression and Random Forest Regressor was conducted. Although Random Forest achieved slightly improved predictive accuracy, Linear Regression was selected for its transparency, lower complexity, and interpretability.

The results indicate that machine learning–based demand forecasting combined with waste analytics can enhance operational decision-making and contribute to sustainability in buffet restaurants. The prototype provides a scalable foundation for further development using real-world restaurant data.

Keywords: Machine Learning, Demand Forecasting, Food Waste, Regression Analysis, Decision Support System

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1 INTRODUCTION

1.1 Background

Food waste is a critical global issue affecting economic sustainability and environmental responsibility. In buffet-style restaurants, food must be prepared in advance without precise knowledge of customer demand. This uncertainty often leads to systematic overproduction. Traditional forecasting approaches rely on human experience, intuition, and historical averages, which may not adequately capture dynamic factors such as temperature changes or holiday effects.

Advances in machine learning enable data-driven demand forecasting, allowing restaurants to shift from intuition-based estimation to predictive analytics. This thesis explores the application of regression-based machine learning techniques to address food demand prediction and waste reduction.

1.2 Problem Statement

Buffet restaurants face the following challenge: How can next-day food demand be predicted accurately while simultaneously reducing food waste?

The problem can be formulated as a supervised regression task where historical operational data is used to predict future demand.

1.3 Objectives

The objectives of this thesis are:

- To understand core machine learning regression principles
- To implement and compare multiple regression models
- To perform feature engineering and data preprocessing
- To evaluate model performance using appropriate metrics
- To integrate waste analytics into the prediction system
- To design a cloud-based decision support architecture

The primary focus of the project is on understanding machine learning methodology rather than purely on coding implementation.

2 THEORETICAL BACKGROUND

2.1 Machine Learning in Demand Forecasting

Demand forecasting is typically formulated as a supervised learning problem. In supervised regression:

- Input features (X) represent influencing factors.

- Target variable (y) represents continuous output (demand).

The model learns a function:

$f(X) \rightarrow y$

In this project:

X = {temperature, holiday indicator, historical demand}

y = next-day demand

2.2 Regression Models

Linear Regression

Linear Regression assumes a linear relationship between independent variables and the target variable:

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \varepsilon$

Where:

- β represents model coefficients
- ε represents error term

Advantages:

- High interpretability
- Low computational complexity
- Easy coefficient interpretation

Limitations:

- Assumes linear relationship
- Limited modelling of complex patterns

Random Forest Regressor

Random Forest is an ensemble learning method combining multiple decision trees.

Advantages:

- Captures nonlinear interactions
- Reduces variance through averaging
- Handles feature interactions automatically

Limitations:

- Reduced interpretability
- Higher computational complexity
- More difficult to explain to stakeholders

3 METHODOLOGY

3.1 Data Preparation

Due to lack of access to real restaurant data, synthetic data was generated simulating daily buffet operations across multiple dishes.

The dataset includes:

- Date
- Dish name
- Temperature (°C)
- Holiday indicator (0/1)
- Cooked quantity
- Sold quantity
- Waste quantity

The dataset covers a full operational year.

3.2 Feature Engineering

Feature preparation included:

1. Temperature as continuous numeric variable
2. Holiday indicator as binary variable
3. Lag demand (previous day demand)
4. Dish encoding for categorical representation

Feature engineering is essential because machine learning models require structured numerical input.

3.3 Model Selection Rationale

Linear Regression was selected as the primary model because:

- Dataset size is moderate
- Interpretability is critical for business stakeholders
- The objective emphasizes conceptual understanding
- Risk of overfitting is lower

Random Forest was implemented to compare performance and assess nonlinear modeling benefits.

3.4 Training and Validation Strategy

The dataset was divided into:

- 80% Training set
- 20% Test set

Temporal ordering was preserved to prevent data leakage.

4 MODEL EVALUATION

4.1 Evaluation Metrics

Mean Absolute Error (MAE)

MAE measures average absolute prediction error: $MAE = (1/n) \Sigma |y_true - y_pred|$

Lower MAE indicates better predictive accuracy.

Root Mean Squared Error (RMSE)

RMSE penalizes larger errors: $RMSE = \sqrt{[(1/n) \Sigma (y_true - y_pred)^2]}$

RMSE is sensitive to large deviations.

4.2 Model Comparison Results

Table 1. Comparison among models

	Model	MAE	RMSE	MAPE (%)
1	Linear Regression	10.20	13.06	9.36
2	Random Forest	10.29	13.60	9.22
0	Naive Baseline	15.69	19.32	14.69

Although Random Forest achieved slightly lower error values, Linear Regression was selected due to:

- Greater transparency
- Easier explanation
- Lower computational cost
- Better alignment with learning objectives

5 SYSTEM IMPLEMENTATION

5.1 Architecture

The system architecture consists of three components:

Google Colab → BigQuery → Looker Studio

Colab handles:

- Data preprocessing
- Model training

- Prediction generation
- Waste computation

BigQuery:

- Stores predictions
- Stores scenario data
- Acts as data warehouse

Looker Studio:

- Interactive dashboard
- Scenario selection
- Visual analytics

6 FOOD WASTE ANALYSIS

Waste is computed using:

$waste_qty = \max(cooked_qty - sold_qty, 0)$

Waste percentage:

$waste_pct = (waste_qty / cooked_qty) \times 100$

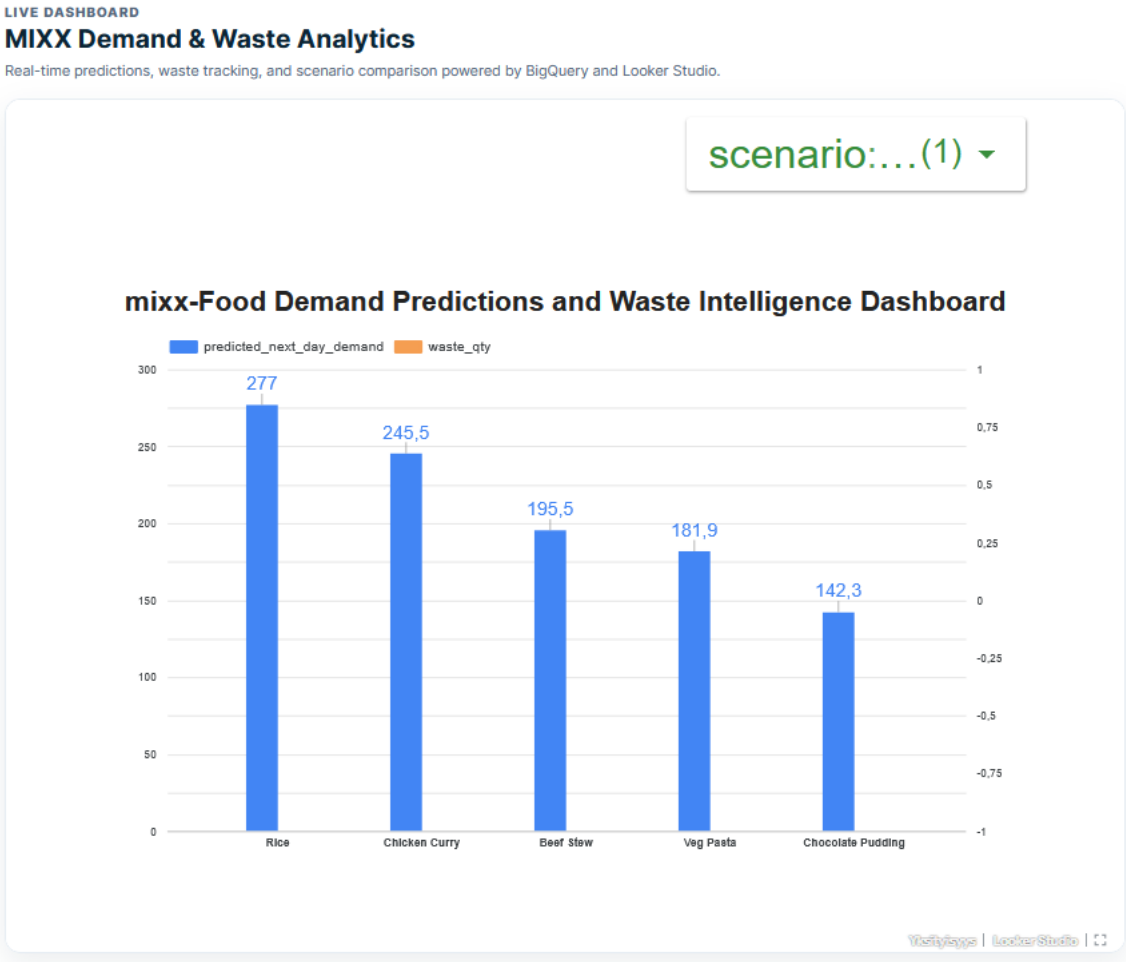
This enables identification of inefficiencies and supports sustainability objectives.

7 DEMONSTRATION SCENARIOS

Demo 1 – Baseline

Historical data only. No operational input. Pure demand forecasting.

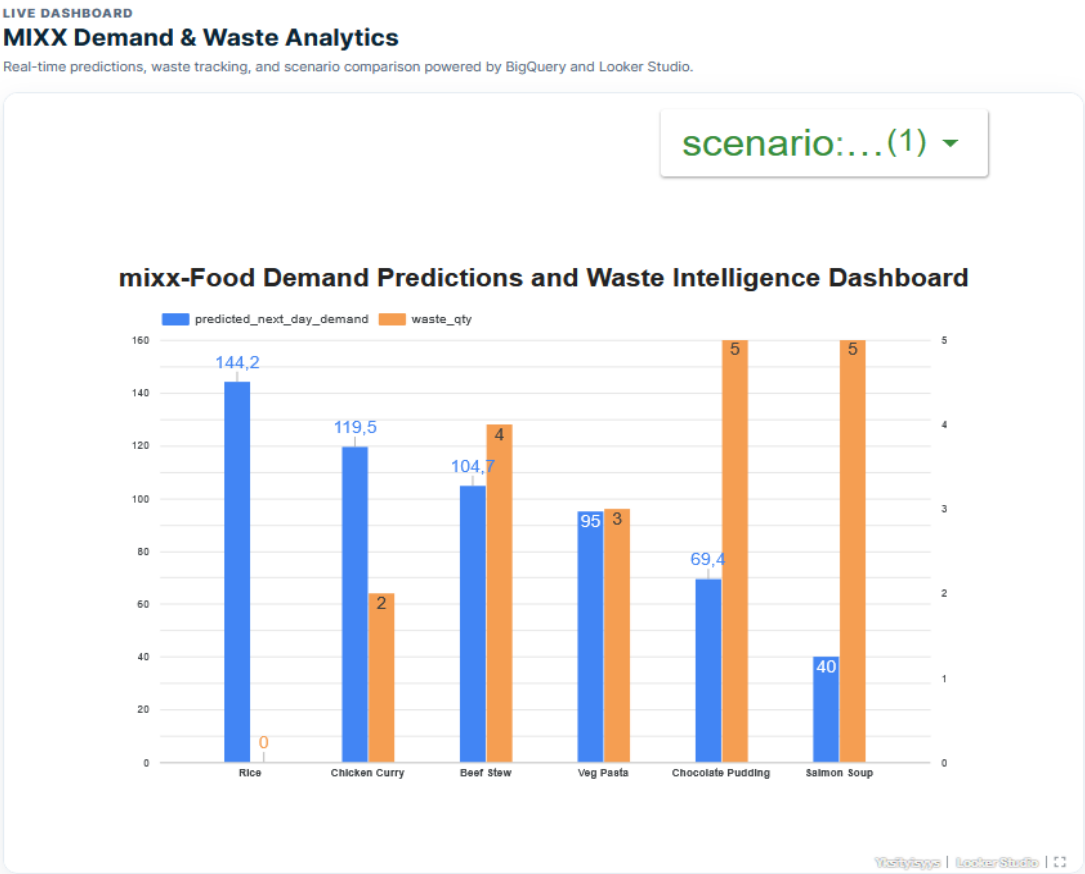
Figure 1. Predict based on historical data.



Demo 2 – Daily Input

Staff inputs sold and cooked quantities. Waste calculated. Model updated.

Figure 2. Input new item and predicting the demand



Demo 3 – New Dish

New dish introduced without historical data. Fallback average used.

This demonstrates robustness and system scalability.

8 RESULTS AND INTERPRETATION

The model achieved acceptable error levels, indicating reasonable predictive accuracy for synthetic data. Temperature and holiday features showed measurable influence on demand.

Waste analytics revealed:

- Some dishes consistently produce higher waste
- Demand variation correlates with external factors
- Data-driven planning can reduce overproduction

The integration of forecasting and waste metrics enhances operational decision-making.

9 DISCUSSION

This project demonstrates understanding of:

- Supervised regression
- Feature engineering
- Model comparison

- Evaluation metrics
- Business-oriented ML application

Limitations:

- Synthetic data
- Limited feature diversity
- Simplified operational assumptions

10 FUTURE WORK

Future extensions include:

- Full MIXX web software development
- POS integration
- Advanced ML models (XGBoost, LSTM)
- Real-time retraining pipelines
- Computer vision waste detection system using smart bin cameras
- Sustainability impact reporting

Integration of computer vision would enable automated waste detection through object recognition models installed above waste bins, significantly improving accuracy.

11 CONCLUSION

This thesis developed and evaluated a machine learning–based food demand prediction and waste analytics system. The study emphasized understanding regression modelling, feature engineering, model evaluation, and result interpretation.

The MIXX prototype demonstrates that predictive analytics combined with waste monitoring can support operational efficiency and sustainability in buffet restaurants. The system provides a scalable framework for future development and real-world deployment.

Appendix 1. Data Set validation

```
Shape: (1305, 10)

Columns: ['date', 'weekday', 'dish_name', 'category', 'price_level', 'temperature_c', 'is_holiday', 'cooked_qty', 'sold_qty', 'wasted_qty']
```

	date	weekday	dish_name	category	price_level	temperature_c	is_holiday	cooked_qty	sold_qty	wasted_qty
0	2025-01-01	Wednesday	Beef Stew	Main	High	17.0	0	98	93	5
1	2025-01-02	Thursday	Beef Stew	Main	High	7.3	0	110	101	9
2	2025-01-03	Friday	Beef Stew	Main	High	26.7	0	115	104	11
3	2025-01-06	Monday	Beef Stew	Main	High	11.9	0	126	117	9
4	2025-01-07	Tuesday	Beef Stew	Main	High	19.4	0	107	99	8
5	2025-01-08	Wednesday	Beef Stew	Main	High	15.2	0	93	89	4
6	2025-01-09	Thursday	Beef Stew	Main	High	3.6	0	108	100	8
7	2025-01-10	Friday	Beef Stew	Main	High	15.6	0	125	114	11
8	2025-01-13	Monday	Beef Stew	Main	High	9.8	0	116	102	14
9	2025-01-14	Tuesday	Beef Stew	Main	High	13.0	0	100	94	6

```
Missing values per column:
date            0
weekday         0
dish_name       0
category        0
price_level     0
temperature_c   0
is_holiday      0
cooked_qty      0
sold_qty        0
wasted_qty      0
dtype: int64

Date range: 2025-01-01 00:00:00 to 2025-12-31 00:00:00
Unique dishes: 5
Unique categories: 4

Target column: sold_qty
Target stats:
```

	sold_qty
count	1305.000000
mean	106.986207
std	28.901405
min	42.000000
25%	85.000000
50%	103.000000

Appendix 2. Dashboard Deployment Link

https://mixx.meherix.com/

Appendix 3. Github Repo Link

https://github.com/jahid-github/Food-Demand-Prediction-and-Waste-Analytics-